26900173 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCECO300 - ECONOMETRICS -I

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. What role does mathematics play in econometric framework? [U] / [CO1]

   a). To automatically gather live stock market prices.

   b). To remove all randomness from society.

   c). To make models impossible to understand.

   d). To translate qualitative economic reasoning into precise testable equations.

2. Linear regression model is: [U] / [CO4]

   a). Linear in explanatory variable but may not be linear in parameters.

   b). Nonlinear in parameters and must be linear in variables.

   c). Linear in parameters and must be linear in variables.

   d). Linear in parameters and may not be linear in variables.

3. What is a hypothesis? [U] / [CO4, CO5]

a). A proven fact about a population

b). A tentative statement or assumption that can be tested statistically

c). A mathematical calculation

d). A sample observation

4. OLS estimators are expressed solely in terms of: [U] / [CO4]

a). Observable quantities of Y and X

b). Unobservable poulation parameters

c). The true error terms

d). Out-of-sample observations

5. The correlation between the error term and the independent variable in multiple regression analysis is equal to [An] / [CO2]

a). 1

b). 0

c). 0-1

d). None of these

6. The error term introduces _____ into an econometric model. [A] / [CO2]

a). Certainty

b). Stochasticity

c). Linearity

d). Determinism

7. Does the Gauss-Markov theorem require the errors to be normally distributed? [A] / [CO4]

a). Yes

b). No

c). Only if N<30

d). Only if there is no intercept

8. Violation of Normality is a more serious problem in: [U] / [CO3]

a). Time series data

b). Large samples

c). Panel data

d). Small samples

9. The ANOVA table helps evaluate: [An] / [CO3, CO5]

a). Overall significance of the regression model

b). Only the median

c). Only the mode

d). Only the sample size

10. What is a Type II error? [U] / [CO4, CO5]

a). Failing to reject a false null hypothesis

b). Rejecting a true null hypothesis

c). Accepting a true null hypothesis

d). Rejecting a true alternative hypothesis

[1x10 = 10]

Part B

Short Answer

Answer any **10** questions

Each question carries **3** marks

11. What is panel data (or longitudinal data), and what makes it unique? [U] / [CO2]

12. What is heteroscedasticity in an econometric model? [U] / [CO2]

13. Explain the scope of Econometrics. [U] / [CO1]

14. What is negative autocorrelation? [U] / [CO2]

15. Illustrate a multiple regression model based on economic theory, showing the effect of two factors on household consumption expenditure. [An] / [CO3]

16. What does a smaller standard error indicate about an estimator? [A] / [CO4]

17. State three assumptions of t-test. [U] / [CO4]

18. Define Sample Regression Function. [K] / [CO4]

19. What is econometric methodology? Briefly explain its main steps [U] / [CO2]

20. "Independent variables should be truly independent in a multiple regression model." Examine the statement. [A] / [CO3]

21. Write down the OLS estimators of the slope and intercept in a simple regression model. [U] / [CO4]

22. Define power of a test. [U] / [CO4, CO5]

23. What does R^2=0.80 mean? [A] / [CO3, CO5]

[3x10 = 30]

Part C

Short Essay

Answer any **5** questions

Each question carries **6** marks

24. State the major assumptions of the Classical Linear Regression Model. [K] / [CO4]

25. Explain how dummy variables are used to represent qualitative characteristics such as gender, religion, and marital status in an econometric model. [A] / [CO2]

26. Illustrate how simple linear regression is utilized in empirical economics by formulating four distinct models grounded in economic theory. [A] / [CO4]

27. Explain how multicollinearity affects standard errors and t-values. [An] / [CO2]

28. Explain the concepts of sampling distribution, point estimation, and interval estimation in linear regression. Discuss the role of standard error in estimating regression coefficients. [An] / [CO4]

29. Explain the importance of the critical region in making a decision about the null hypothesis. [U] / [CO4, CO5]

30. How is R^2 calculated using ESS and TSS? [An] / [CO3, CO5]

[6x5 = 30]